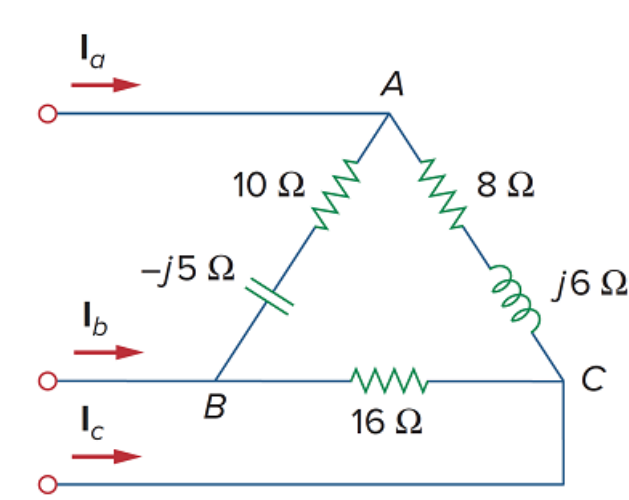


Problem

The unbalanced Δ -load of Fig. 12.24 is supplied by balanced line-to-line voltages of 440 V in the positive sequence. Find the line currents. Take $\mathbf{V}_{ab}$ as reference.

Figure 12.24



Step-by-step solution

Step 1 of 5

Draw the circuit diagram:

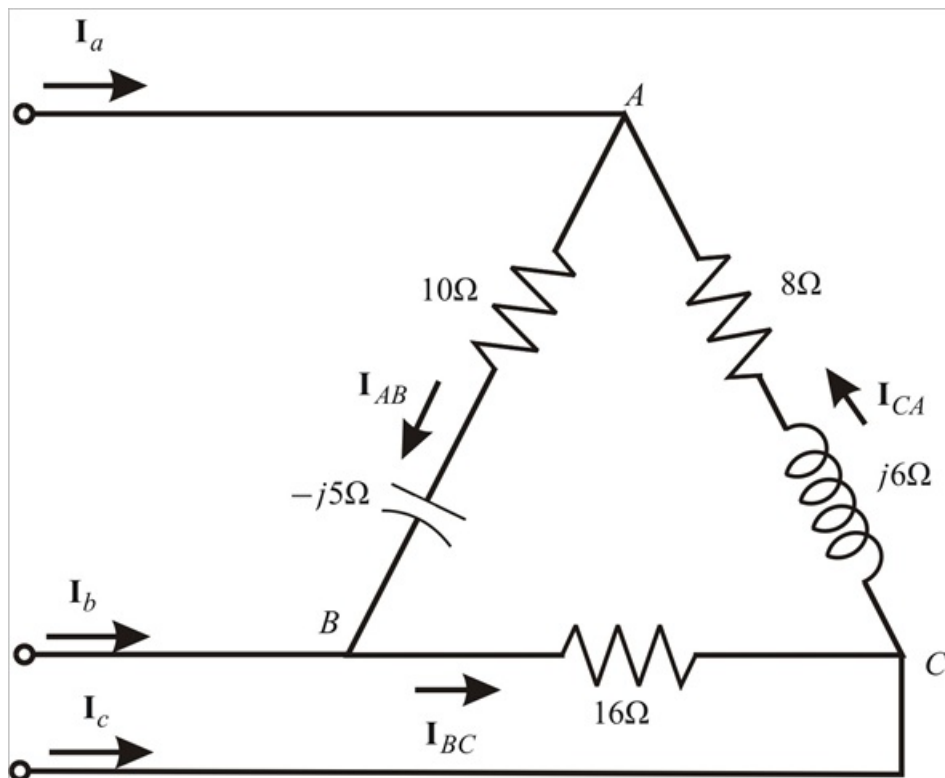


Figure 1

Step 2 of 5

Consider the following data:

Line-to-line voltage is 440 V and positive sequence

Calculate the phase voltage in delta connected load.

$$\mathbf{V}_{AB} = 440\angle 0^\circ \text{ V}$$

$$\begin{aligned}\mathbf{V}_{BC} &= \mathbf{V}_{AB}\angle -120^\circ \\ &= 440\angle (0^\circ - 120^\circ) \text{ V} \\ &= 440\angle -120^\circ \text{ V}\end{aligned}$$

$$\begin{aligned}\mathbf{V}_{CA} &= \mathbf{V}_{AB}\angle 120^\circ \\ &= 440\angle (0^\circ + 120^\circ) \text{ V} \\ &= 440\angle 120^\circ \text{ V}\end{aligned}$$

Step 3 of 5

Calculate the phase currents.

$$\begin{aligned}\mathbf{I}_{AB} &= \frac{\mathbf{V}_{AB}}{\mathbf{Z}_{AB}} \\ &= \frac{440\angle 0^\circ}{10 - j5} \\ &= \frac{440\angle 0^\circ}{11.18\angle -26.56^\circ} \\ &= 39.35\angle 26.56^\circ \text{ V}\end{aligned}$$

$$\begin{aligned}\mathbf{I}_{BC} &= \frac{\mathbf{V}_{BC}}{\mathbf{Z}_{BC}} \\ &= \frac{440\angle -120^\circ}{16} \\ &= 27.5\angle -120^\circ \text{ A}\end{aligned}$$

$$\begin{aligned}\mathbf{I}_{CA} &= \frac{\mathbf{V}_{CA}}{\mathbf{Z}_{CA}} \\ &= \frac{440\angle 120^\circ}{8 + j6} \\ &= \frac{440\angle 120^\circ}{10\angle 36.86^\circ} \\ &= 44\angle 83.12^\circ \text{ A}\end{aligned}$$

Step 4 of 5

Apply Kirchhoff's current law at node 'A', to obtain the line current $\mathbf{I}_a$.

$$\mathbf{I}_a + \mathbf{I}_{CA} = \mathbf{I}_{AB}$$

$$\begin{aligned}\mathbf{I}_a &= \mathbf{I}_{AB} - \mathbf{I}_{CA} \\ &= 39.35\angle 26.56^\circ - 44\angle 83.12^\circ \\ &= 35.19 + j17.59 - 5.27 - j43.16 \\ &= 29.926 - j26.08\end{aligned}$$

$$= 39.7\angle 41.08^\circ \text{ A}$$

Therefore, the line current $\mathbf{I}_a$ is $\boxed{\mathbf{I}_a = 39.7\angle 41.08^\circ \text{ A}}$.

Step 5 of 5

Apply Kirchhoff's current law at node 'B', to obtain the line current $\mathbf{I}_b$.

$$\mathbf{I}_b + \mathbf{I}_{AB} = \mathbf{I}_{BC}$$

$$\begin{aligned}\mathbf{I}_b &= \mathbf{I}_{BC} - \mathbf{I}_{AB} \\ &= 27.5\angle -120^\circ - 39.35\angle 26.56^\circ \\ &= -13.75 - j23.81 - 35.19 - j17.59 \\ &= -48.94 - j41.74\end{aligned}$$

$$= 64.11\angle -139.76^\circ \text{ A}$$

Therefore, the line current $\mathbf{I}_b$ is $\boxed{\mathbf{I}_b = 64.11\angle -139.76^\circ \text{ A}}$.

Apply Kirchhoff's current law at node 'B', to obtain the line current $\mathbf{I}_c$.

$$\mathbf{I}_c + \mathbf{I}_{BC} = \mathbf{I}_{CA}$$

$$\begin{aligned}\mathbf{I}_c &= \mathbf{I}_{CA} - \mathbf{I}_{BC} \\ &= 44\angle 83.12^\circ - 27.5\angle -120^\circ \\ &= 5.27 + j43.16 + 13.75 + j23.81 \\ &= 19.02 + j67.49\end{aligned}$$

$$= 70.12\angle 74.26^\circ \text{ A}$$

Therefore, the line current $\mathbf{I}_c$ is $\boxed{\mathbf{I}_c = 70.12\angle 74.26^\circ \text{ A}}$.